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Capability Surprise as the Pricing Mechanism of a Pre-Revenue Foundation-Model Lab: How Model-Release and Benchmark Events Drive Price Discovery and Drift (Zhipu AI, 2513.HK)

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Abstract

In January 2026 Zhipu AI (Knowledge Atlas Technology, 2513.HK) became the first foundation-model laboratory to list publicly anywhere, debuting on the Hong Kong main board under the Chapter 18C regime for pre-revenue specialist-technology firms. Within five months its shares rose roughly nineteen-fold, lifting market capitalisation to about US\$124 billion on expected 2026 revenue of only US\$200 million—an implied price-to-sales multiple near $620\times$. This paper asks what such a company is worth and how its price is formed. We proceed on two tracks. First, we build a fundamental anchor: a CAPM cost of capital, a three-scenario discounted-cash-flow (DCF) model, an EV/Sales comparison with peer MiniMax and global laboratories, and a real-options reading. Even generous assumptions—revenue compounding 21–43% annually for a decade—yield an intrinsic value of only HK\$29–84 per share against a market price near HK\$2,228; a reverse DCF shows the price embeds roughly 90% annual revenue growth and US\$64 billion of revenue by 2035, hyperscaler-tier expectations. Second, because a pre-revenue lab has no earnings to surprise, we replace the classic earnings-surprise event study with a *capability-surprise* one: treating model releases and benchmark-leaderboard jumps (GLM-5 through GLM-5.2) as information events and measuring cumulative abnormal returns. The market reacts immediately and discriminately—separating Zhipu from MiniMax on model quality rather than sector—but mis-times magnitude, under-reacting to genuine state-of-the-art leaps and over-reacting to incremental ones, evidence of post-capability-announcement drift that is robust to peer adjustment. We conclude that the valuation gap is best read not as a simple bubble but as a market pricing a winner-take-all option on frontier AI, whose path is driven by capability momentum.

Keywords: foundation models; Chapter 18C; pre-revenue valuation; real options; market efficiency; event study; PEAD/PCAD.

1 Introduction

The first half of 2026 produced an event with no precedent in capital-market history: within forty-eight hours, two Chinese “AI tiger” laboratories—Zhipu AI (2513.HK, 8 January) and MiniMax (00100.HK, 9 January)—listed on the Hong Kong Stock Exchange, the first foundation-model developers to go public anywhere.¹ Zhipu listed under Chapter 18C, a regime designed to admit pre-revenue specialist-technology firms. What followed was extraordinary even by the standards of growth investing: Zhipu’s shares rose roughly nineteen-fold in five months to a market capitalisation near US\$124 billion,² on FY2026 expected revenue of about US\$200 million.

This paper poses three questions. **(i)** What is Zhipu worth on fundamentals? **(ii)** What does the nineteen-fold re-rating imply—efficient pricing of capability, exuberance, or option value? **(iii)** Is the open-weight, cost-disruption strategy that drives its leaderboard dominance financially sustainable? Our contribution is methodological as much as empirical. Because a pre-revenue

¹CNBC, “The first of China’s AI tigers goes public” (2026-01-08); “MiniMax doubles in Hong Kong debut” (2026-01-09).

²Price and market-capitalisation data: stockanalysis.com (HKG:2513), retrieved 2026-06-26; daily series from Tushare `hk_daily`.

lab has no earnings, the standard apparatus of market-efficiency testing—the earnings-surprise event study [Ball and Brown, 1968, Bernard and Thomas, 1989]—has nothing to bite on. We therefore reframe the information event as a *capability surprise*: a model release or a benchmark-leaderboard jump. This adapts the “earnings-surprise” (*chao yuqi*) toolkit of Guosen Securities Economic Research Institute [2020] to a new asset class, reusing its cumulative-abnormal-return (CAR) windows and a deliberately diagnostic, small-sample-honest posture.

2 Company and Business Model

Zhipu AI was founded in 2019 out of Tsinghua University’s Knowledge Engineering Group and renamed Knowledge Atlas Technology ahead of listing. It develops the GLM (“General Language Model”) family and monetises through three channels: usage-based API access, enterprise subscriptions, and private/on-premise model deployment plus fine-tuning. Its distinctive strategic choice is an *open-weight funnel*: flagship models (GLM-4.6, GLM-5.x) are released with open weights and aggressive token pricing, converting developer mindshare into a usage and data flywheel routed through the Z.ai and Z.code developer platforms. Roughly 70% of IPO proceeds are earmarked for model R&D over 2026–2028.³

3 Financial Performance

Zhipu’s reported financials are those of an early-stage laboratory, not a mature firm, and three features dominate (Section A gives the full statements). **(1) Explosive but tiny revenue:** revenue rose from RMB 312.4m (2024) to RMB 724.3m (2025, +132%), roughly US\$44m then US\$102m. **(2) A collapsing gross margin:** gross margin fell from 76.1% (2022)—a classic software profile—to 3.4% (2024) and to −0.4% in H1 2025, as the firm bought market share through compute-heavy, under-priced API and training workloads. **(3) Deepening losses and negative equity:** the net loss widened to about RMB 4.70bn in 2025 (+237%); at 30 June 2025 the group reported net liabilities of RMB 6.15bn and net current liabilities of RMB 9.56bn against cash of only RMB 190.9m.⁴ Conventional ratio analysis is therefore of limited use: with negative margins and negative book equity, the informative metrics are revenue growth, the gross-margin trajectory, cash burn, and post-IPO runway. We take FY2026 revenue of US\$200m as the launch point for the valuation in Section 8.

4 Market Reaction and Capital-Market Strategy

Zhipu priced its IPO at HK\$116.20, raising about HK\$4.3bn with HK\$3bn of cornerstone commitments, at an implied valuation near US\$6.7–7.4bn.⁵ Three features make the listing strategically remarkable. First, **regulatory access:** Chapter 18C let a loss-making lab tap permanent public capital years before profitability, ending dependence on private rounds. Second, **certification:** being the world’s first listed foundation-model company conferred a signalling premium. Third, **the re-rating:** the stock compounded to roughly HK\$2,228 by late June 2026 (Figure 1), about +1,920% versus IPO, with realised annualised volatility near 192%. MiniMax, listing one day later at HK\$165, doubled on debut, then *de-rated* by roughly 60% from its peak—a divergence we exploit in Section 9.

³Caixin Global, “Zhipu AI Seeks Up to \$640 Million in Hong Kong IPO” (2025-12-30).

⁴Prospectus and interim figures: HKEXnews filing for stock code 2513 (2025-12-30); summary via Tiger Brokers and Simply Wall St.

⁵Baker McKenzie, “Advises Knowledge Atlas Technology on its HKD 4.3 Billion Global Offering” (2026-01); Caixin Global (2026-01-08).

5 Industry and Competitive Landscape

A Porter’s five-forces reading of foundation models is unusual: supplier power (advanced compute) is high and partly politically rationed; buyer switching costs are low because open weights commoditise the layer; and the credible substitutes are other open-weight frontier models. In this structure Zhipu’s leaderboard strategy is the competitive core. Its GLM-5 line—built on a mixture-of-experts architecture (GLM-4.6: $\approx 355\text{B}$ parameters, 32B active, 200K context)—topped coding benchmarks: GLM-5.1 ranked #1 on SWE-Bench Pro at 58.4%, within 2.6 points of the leading closed model, and GLM-5.2 (June 2026) shipped with MIT open weights and a one-million-token context at unchanged pricing.⁶ The strategic logic is *cost disruption*—give away capability at the frontier and price tokens below closed competitors—and it is now paired with emerging *pricing power*: each GLM-5.x release was accompanied by API price increases of 8–17%.⁷ This release cadence defines the event taxonomy for Section 9.

6 Macroeconomic and Policy Environment

Three macro forces dominate. **Industrial policy**: Chinese AI is a national priority, supporting demand through state and SOE procurement. **Compute export controls**: U.S. restrictions on advanced chips force architectural efficiency (MoE, aggressive quantisation), an “efficiency-by-necessity” that doubles as a cost advantage. **Capital-market structure**: Hong Kong’s Chapter 18C, Hang Seng Tech Index inclusion, and Southbound Stock Connect eligibility jointly determine the marginal buyer—Zhipu joined the Hang Seng Tech Index on 5 June and Stock Connect on 8 June 2026, drawing about HK\$920m of net southbound buying on its first eligible day.⁸ With the HKD peg and the global rate cycle, small changes in risk appetite move such a long-duration, optionality-laden asset sharply.

7 Corporate Strategy and SWOT

Zhipu’s strategy is an open-source flywheel layered over enterprise, sovereign, and overseas expansion. **Strengths**: frontier capability, open-weight mindshare, first-mover capital access. **Weaknesses**: deep losses, near-zero gross margin, compute dependence. **Opportunities**: agentic/coding workloads, sovereign AI demand, international developers, nascent pricing power. **Threats**: a brutal release cadence that commoditises its own product, chip access, and valuation air. These drivers parameterise the scenarios next.

8 Valuation

8.1 Cost of capital

We estimate the cost of equity with the CAPM. Lacking a usable local index series, we relevé a bottom-up beta from global AI-software comparables, giving $\beta \approx 1.6$. With a risk-free rate of 4.0% and an equity risk premium of 6.0%, $K_e = 4.0\% + 1.6 \times 6.0\% = 13.6\%$. Because the firm is effectively all-equity, we take WACC $\approx 13.5\%$.

⁶Model specs/benchmarks: howaiworks.ai (GLM-4.6), serenitiesai.com (GLM-5.1), eigent.ai (GLM-5.2).

⁷Investing.com, “Zhipu shares jump on second AI price hike this year” (2026-04-08).

⁸Futu/India Infoline, Hang Seng Tech inclusion and Stock Connect flows (2026-06).

8.2 Scenario DCF

We project FCFF over 2026–2035 from a US\$200m base, with revenue growth fading from a scenario-specific start to a mature rate, operating margin ramping from -30% to a terminal level, a 15% tax rate, and reinvestment via a sales-to-capital ratio of 1.6–2.0; terminal value uses $g = 4\%$. All cells are live in the companion workbook (`model/valuation_model.xlsx`); results are in Table 1.

Table 1: Three-scenario DCF (USD unless noted; per-share in HKD).

Scenario	Rev. CAGR '26–35	Term. margin	Equity value	Per share (HK\$)
Bear ($p = 0.35$)	21%	18%	US\$1.6B	29
Base ($p = 0.45$)	31%	28%	US\$2.5B	46
Bull ($p = 0.20$)	43%	35%	US\$4.7B	84
Probability-weighted	—	—	US\$2.6B	47
<i>Market (Jun 2026)</i>	—	—	<i>US\$124B</i>	<i>2,228</i>

The probability-weighted intrinsic value is about HK\$47 per share—roughly **2% of the market price**. No reasonable point estimate of fundamentals approaches the quoted price.

8.3 Reverse DCF: what the price requires

We invert the model to ask what assumptions justify the US\$124bn equity value. Holding WACC at 13.5% and a 40% terminal margin, the price requires initial revenue growth near **200% per year** fading to 15%, implying revenue of about **US\$64bn by 2035** (a 90% ten-year CAGR, $\approx 318 \times$ FY2026E). This is hyperscaler-tier scale—an outcome reserved for a handful of global winners.

8.4 Relative valuation

On FY2026E revenue, Zhipu trades near **620 $\times$ sales**. For context, leading private frontier labs (OpenAI, Anthropic) have transacted around 30–40 $\times$ trailing revenue, and MiniMax trades far lower than Zhipu on the same basis. Zhipu is thus priced roughly an order of magnitude above even the most richly valued private peers—consistent with the reverse-DCF reading rather than with comparable multiples.

8.5 A real-options reading

The reconciliation is that Zhipu is best valued not as a discounted cash-flow stream but as a *call option on a winner-take-all AGI outcome* [Black and Scholes, 1973]. The DCF base case is the option’s modest intrinsic value; the bulk of the market price is time value on a fat right tail—high volatility (192% annualised) and a long horizon both inflate option value. This framing explains why the stock can rationally trade far above any DCF point estimate while remaining acutely sensitive to capability news—the subject of Section 9.

9 Capability Surprise: An Event Study

Hypothesis. If a pre-revenue lab is priced on its capability trajectory, then capability information—model releases and benchmark-leaderboard jumps—should move the price. Under semi-strong market efficiency the reaction should be immediate and complete; any systematic *post-capability-announcement drift* (PCAD)—the analogue of the post-earnings drift of Ball and Brown [1968], Bernard and Thomas [1989]—would indicate inefficiency.

Method. On authoritative daily prices (Tushare `hk_daily`), we compute mean-adjusted abnormal returns and CARs over three windows: leakage $[-5, -1]$, reaction $[0, +1]$, and drift $[+2, +10]$, following the surprise-event design of Guosen Securities Economic Research Institute [2020]. We first detected Zhipu’s largest abnormal up-days *blind*; they coincide almost one-for-one with GLM releases (Figure 2), validating the event dates against price action rather than press dates.

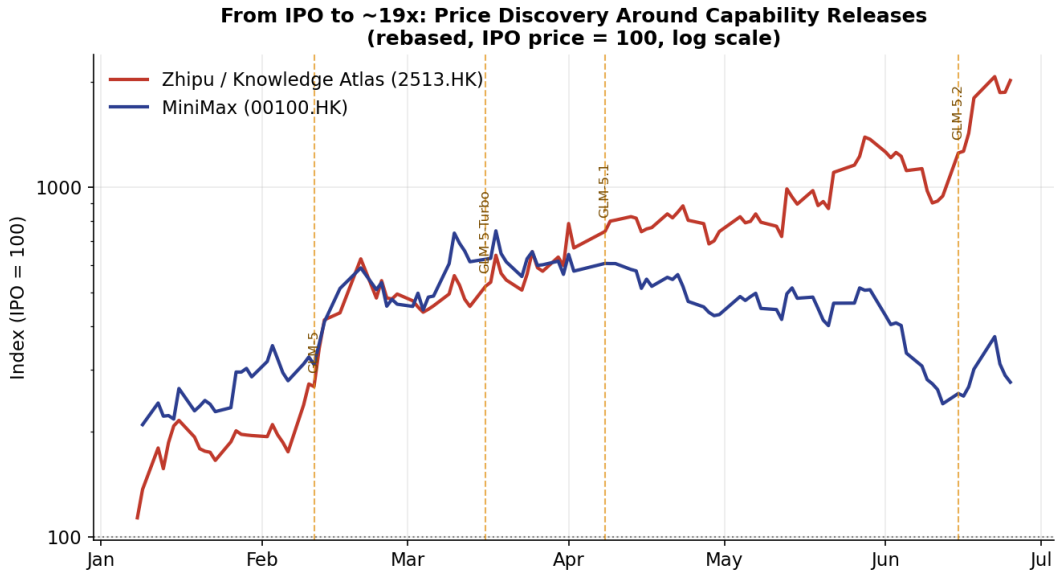


Figure 1: Rebased prices (IPO = 100, log scale) with GLM release markers. Same sector, divergent paths: Zhipu re-rates on successive SOTA releases while MiniMax de-rates from its peak.

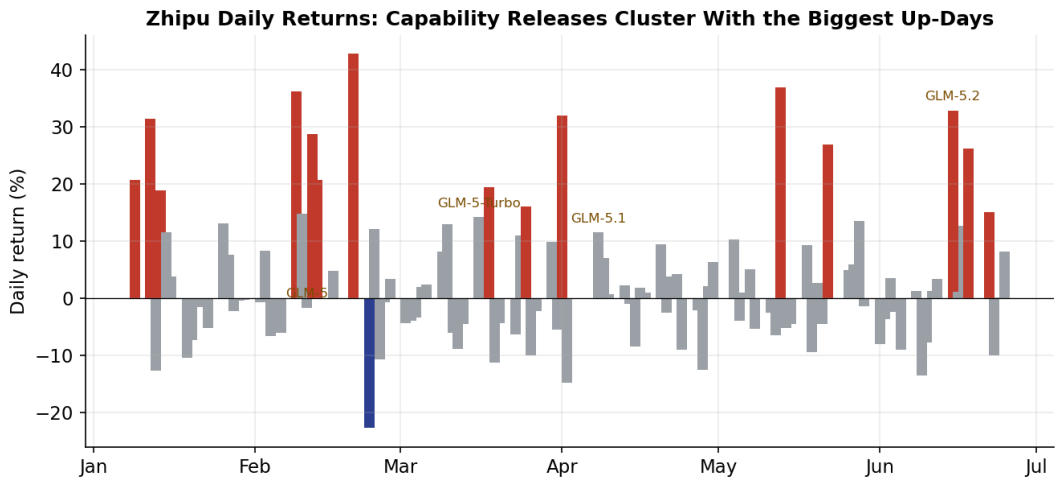


Figure 2: Zhipu daily returns. The largest up-days cluster on capability releases—the market trades a “capability calendar.”

Results. Averaged across the four GLM events, leakage is -6.8% (no systematic front-running), the reaction window is $+17.5\%$ (the market prices capability, fast), and drift is $+4.3\%$ on average but strongly *bimodal* (Table 2, Figures 3–4). The drift sign depends on the *type* of release: a genuine SOTA leap (GLM-5.2) keeps re-rating ($+32\%$ drift, under-reaction), whereas an incremental upgrade (GLM-5.1) reverses (-23% drift, over-reaction). MiniMax’s M2.7 produced a muted reaction and the stock subsequently de-rated, confirming that *capability surprise, not sector membership, drives the cross-section*.

Table 2: Cumulative abnormal returns around capability events (mean-adjusted, %).

Event	Day 0	Reaction [0, +1]	Drift [+2, +10]	Reading
GLM-5	2026-02-11	+20.4	+15.4	under-reaction
GLM-5-Turbo	2026-03-16	+10.3	-7.7	mild reversal
GLM-5.1	2026-04-08	+11.9	-22.8	over-reaction
GLM-5.2	2026-06-15	+27.4	+32.4	strong under-reaction
<i>MiniMax M2.7</i>	2026-03-18	+4.5	-3.8	muted / de-rate

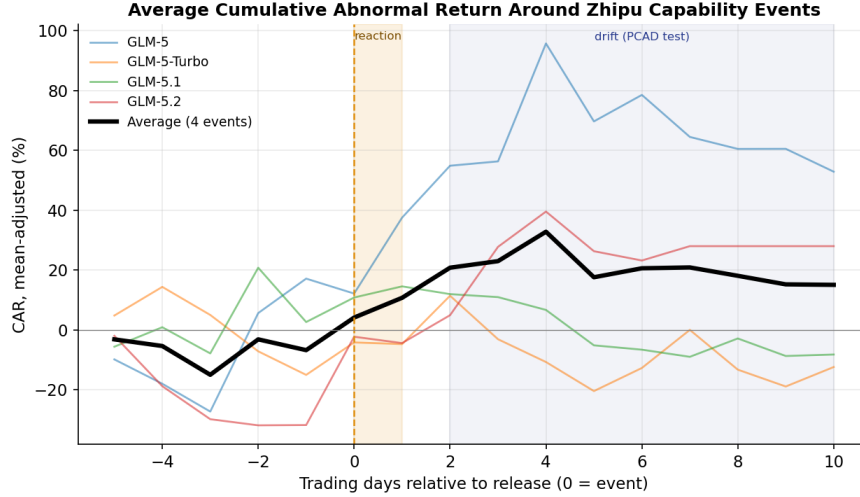


Figure 3: Average CAR in event time across the four GLM events; the average keeps rising after the event (positive drift / PCAD).

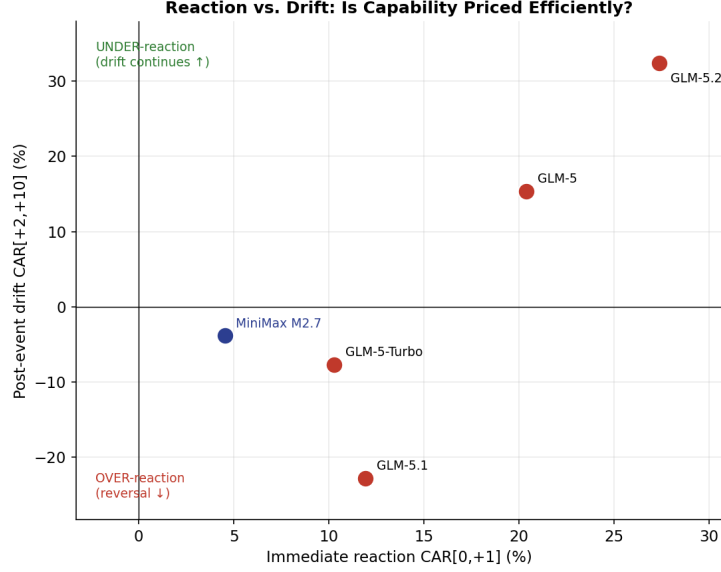


Figure 4: Reaction versus drift. SOTA leaps sit in the under-reaction quadrant; incremental upgrades in the over-reaction quadrant.

9.1 Not every spike is capability: flow events

Two of the largest blind-detected up-days do *not* map to a model release: +43% on 2026-02-20 and +37% on 2026-05-13. Both coincide with *index/flow* catalysts rather than capability—rising expectations of Hang Seng Tech Index inclusion and Southbound Stock Connect access (later realised on 5 and 8 June, with HK\$920m of first-day southbound buying). Separating capability events from flow events matters: the CAR results above are driven by the former, while the latter explain part of the raw volatility and warn against attributing every move to model quality.

9.2 Robustness: peer-adjusted abnormal returns

Because Zhipu and MiniMax share sector and macro shocks, we re-run the study using *peer-adjusted* abnormal returns (benchmark = MiniMax’s same-day return), which strips out the common AI-sector component. Table 3 shows the reaction result is highly robust (+17.5% vs +18.5%), while the average drift *strengthens* (+4.3% → +20.8%): once the falling sector is removed, all four events show positive drift. Notably GLM-5.1’s apparent “over-reaction” flips to continuation (−22.8% → +12.9%), revealing that its mean-adjusted reversal was largely a sector move, not firm-specific. The PCAD/under-reaction conclusion is thus reinforced, not weakened, by peer adjustment.

Table 3: Robustness: mean-adjusted vs peer-adjusted CAR (%).

Event	Mean-adjusted		Peer-adjusted	
	React [0, 1]	Drift [2, 10]	React [0, 1]	Drift [2, 10]
GLM-5	+20.4	+15.4	+17.2	+13.3
GLM-5-Turbo	+10.3	−7.7	+14.6	+19.2
GLM-5.1	+11.9	−22.8	+13.4	+12.9
GLM-5.2	+27.4	+32.4	+28.7	+37.7
Average	+17.5	+4.3	+18.5	+20.8

Honesty box. This is diagnostic, not definitive: $n = 4$ single-firm events (plus one peer) over five months, with drift windows that can overlap a fast release cadence. No statistical significance is claimed. Remaining work: a wider, multi-lab event panel and an NLP-scored surprise magnitude.

10 Conclusion and Implications

A disciplined valuation cannot reconcile Zhipu’s fundamentals (HK\$29–84 per share) with its market price (HK\$2,228). The reverse DCF shows why: the price embeds a US\$64bn-revenue, winner-take-all outcome, best expressed as a call option on frontier AI rather than as a cash-flow stream. The event study shows *how* that price is discovered—through capability surprises that the market prices immediately and discriminately, yet under-reacts to in magnitude (a PCAD that survives peer adjustment), with separate index/flow events driving the rest. The verdict is neither “efficient” nor “bubble” but **capability momentum priced as an option**. For investors, this argues for trading the capability calendar with attention to release *type*; for founders, Chapter 18C has turned capability milestones into the primary currency of value; for policymakers, compute access is now a direct input to capital formation. Limitations are the small event sample and a near-zero gross margin whose future path is highly uncertain; the natural extensions are a multi-lab panel and a text-based capability-surprise score.

A Financial Statements Summary

Figures from the Chapter 18C prospectus and the H1 2025 interim disclosure (RMB; USD at ≈ 7.1).

Table 4: Income-statement summary (RMB millions unless noted).

	FY2022	FY2023	FY2024	FY2025
Revenue	—	≈ 120	312.4	724.3
growth	—	—	—	+132%
Gross margin	76.1%	—	3.4%	$\approx 0\%$ (H1: -0.4%)
Net loss	—	—	2,958.0	$\approx 4,700$
loss growth	—	—	—	+237%

Table 5: Balance-sheet highlights (RMB millions, at 30 June 2025).

Item	Amount
Cash and cash equivalents	190.9
Net liabilities (negative equity)	$-6,150.8$
Net current liabilities	$-9,564.8$
Shares outstanding (post-IPO, approx.)	$\approx 435\text{m}$
IPO price / gross proceeds	HK\$116.20 / $\approx$ HK\$4.3bn

The gross-margin collapse from 76% to near zero is the central operating fact: Zhipu is converting a high-margin software model into a compute-subsidised, share-grab model. Combined with negative equity, it is why the valuation in Section 8 must rest on a future-option logic rather than on current fundamentals.

References

- Ray Ball and Philip Brown. An empirical evaluation of accounting income numbers. *Journal of Accounting Research*, 6(2):159–178, 1968.
- Victor L. Bernard and Jacob K. Thomas. Post-earnings-announcement drift: Delayed price response or risk premium? *Journal of Accounting Research*, 27:1–36, 1989.
- Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973.
- Guosen Securities Economic Research Institute. 金融工程专题研究：超预期投资全攻略 [special report on financial engineering: A complete guide to earnings-surprise investing]. Technical report, Guosen Securities, Sep 2020.